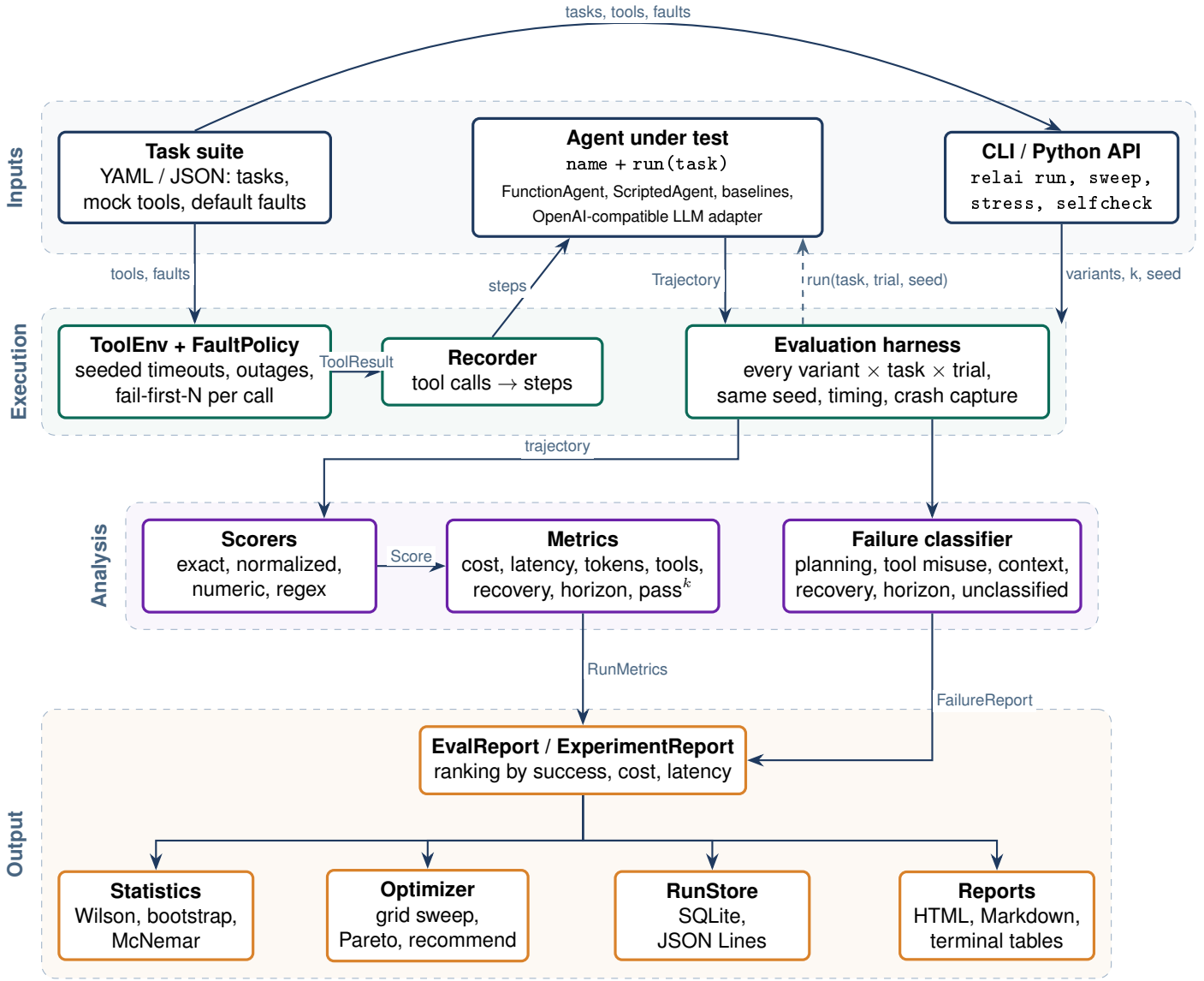
tasks, tools, faults
Inputs
Task suite
YAML / JSON: tasks,
mock tools, default faults
Agent under test
name + run(task)
FunctionAgent, ScriptedAgent, baselines,
OpenAI-compatible LLM adapter
CLI / Python API
relai run, sweep,
stress, selfcheck
tools, faults
steps
Trajectory
run(task, trial, seed)
variants, k, seed
Execution
ToolEnv + FaultPolicy
seeded timeouts, outages,
fail-first-N per call
ToolResult
Recorder
tool calls → steps
Evaluation harness
every variant × task × trial,
same seed, timing, crash capture
trajectory
Analysis
Scorers
exact, normalized,
numeric, regex
Score
Metrics
cost, latency, tokens, tools,
recovery, horizon, pass^k
Failure classifier
planning, tool misuse, context,
recovery, horizon, unclassified
RunMetrics
FailureReport
Output
EvalReport / ExperimentReport
ranking by success, cost, latency
Statistics
Wilson, bootstrap,
McNemar
Optimizer
grid sweep,
Pareto, recommend
RunStore
SQLite,
JSON Lines
Reports
HTML, Markdown,
terminal tables